

Data Preparation and Model Exploration

Retail Location Intelligence | Café site selection in Çankaya, Ankara

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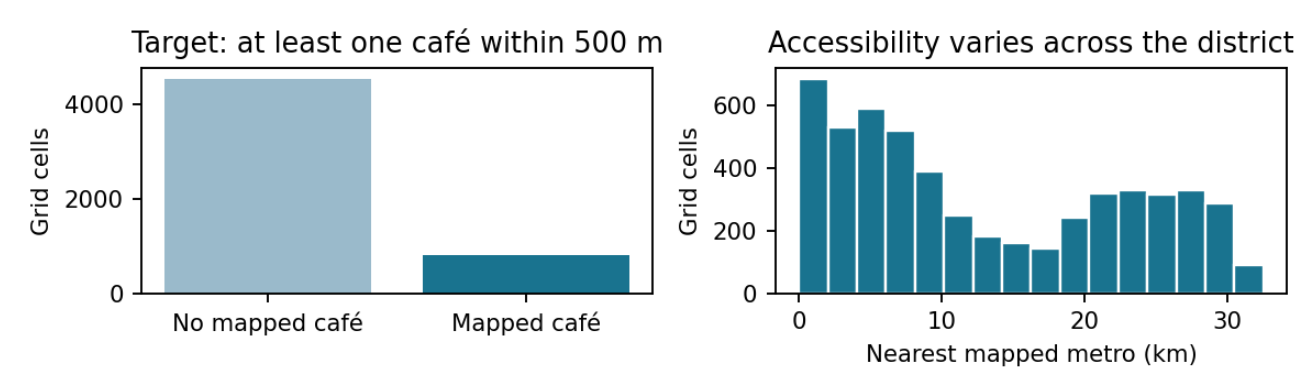
1. Overview, collection and cleaning

The pipeline converts open place data and aggregated residential population into comparable cell-level features for café market planning. OSMnx downloads POIs and drive/walk networks for Çankaya; tagged amenities are classified into exclusive layers. Empty geometries are removed, coordinates are projected to EPSG:32636, polygon POIs are represented by centroids, and list-valued tags are flattened for storage [1, 2].

The district is clipped into 300 m cells. Missing raw count-layer files yield missing feature values; a present empty layer yields zero counts. Schools require explicit ISCED 2/3 tags on refresh. Population is allocated from neighborhood totals by intersection-area share; unmatched neighborhoods can receive median-density estimates. The CSV has 123 rows labeled 2025; all youth-population entries are missing [3, 4].

2. Exploratory analysis

The committed GeoParquet contains 5,361 cells, 121 neighborhood labels and no duplicate cell IDs. There are 817 café-positive buffers (15.24%). Median café, restaurant and shop counts are zero; maxima are 49, 48 and 104 respectively. High counts are retained as possible real urban clusters; no statistical outlier-removal routine is implemented.



Median metro distance is 10.0 km; median residential density is 82.35/km² versus a mean of 2,057.51/km², indicating substantial spatial heterogeneity. Street names are missing in 63.7% of cells, and age-group values are entirely missing. Existing data lacks the newer office, government, kindergarten and nested-competition columns; updated school rules are not retroactively present in this snapshot.

3. Feature engineering

Feature family	Construction / rationale
Demand	University 1 km; school 750 m; shop, park, office, government and kindergarten 500 m. Proxies for nearby activity; cafés and restaurants are excluded from MCDA demand.
Access / population	Bus stops 400 m; nearest metro; intersections per cell; area-weighted residential population and density. Euclidean proximity is not actual walking time.
Competitive pressure	Restaurants/fast food 500 m; distinct cafés in central, 3×3 and 5×5 cells. Thresholds ≥2 / ≥6 / ≥13 define competition warnings.
RF target / predictors	Target: ≥1 café within 500 m. Twelve predictors include café-free diversity, restaurants, retail, transit and population; direct café counts are excluded.

4. Transformation and scoring

MCDAs numeric layers use district-wide MinMax scaling, with median filling and zero fallback; missing demand layers are omitted and their available weights are rescaled. Metro proximity is $1/(1+\text{distance})$, with missing distances treated as far. Saturation preserves constant positive counts as full pressure. Missing penalties remain visible and make scores provisional. RF predictors are not MinMax-scaled; only training-fold median imputation is used.

```
# Actual production scoring: positive weights normalize.
from src.scoring import score_grid
ranked = score_grid(grid)
# Components below are on the 0-100 display scale:
# S = clip((.45*D + .25*A + .20*P)/.90
#         - .03*Sat - .07*Comp, 0, 100)
```

Demand subweights: universities .30, shops .25, schools .15, parks .10, offices .10, government .05, kindergartens .05. Access subweights: bus stops .45, metro proximity .35, intersections .20. These are user-selected scenarios, not learned commercial coefficients. Popularity compatibility is disabled due to absent independent data.

5. Model selection and training

Random Forest provides a nonlinear tabular comparison layer and descriptive feature importance [5]. It is used alongside the primary MCDA score. Logistic regression could be a simpler baseline, but no comparative training experiment is recorded. RF uses 300 trees, depth 8, minimum leaf size 5, balanced class weights and seed 42. Five-fold StratifiedGroupKFold holds out complete neighborhoods [6].

```
from src.build_features import build_features
from src.cafe_similarity import train_and_score

# Requires downloaded raw layers; rebuilds spatial features.
grid = build_features()
scored, cv_report = train_and_score(grid)
# Displayed similarity = 100 * out-of-fold probability.
```

6. Evaluation and marketing interpretation

The recorded corrected snapshot run gives fold ROC-AUC values 0.990, 0.900, 0.953, 0.972 and 0.985 (mean 0.960) [4]. No confusion matrix or ROC curve is recorded, and no threshold or independent test set is validated. Neighborhood grouping reduces leakage, but overlapping cross-border buffers remain a limitation. The result indicates similarity to mapped café presence, not business success. The proposed marketing KPI is shortlist preparation time in a future user pilot.

References and implementation evidence

- [1] OpenStreetMap contributors, copyright and ODbL attribution. <https://www.openstreetmap.org/copyright>
- [2] Boeing, G. (2025), Modeling and Analyzing Urban Networks and Amenities with OSMnx; package documentation. <https://osmnx.readthedocs.io/en/stable/>
- [3] TÜİK, ADNKS population portal; repository CSV labels its observations as 2025. <https://nip.tuik.gov.tr/Home/Adnks>
- [4] Group 6, current main implementation (src/build_features.py, src/allocate_population.py, src/scoring.py, docs/model_validation.md); reviewed 6 October 2026. https://github.com/edasaruhan/SIC_AI_17_Capstone_Group_6/tree/58ce550c027d3ddce9f209fe3979bea849d5716a
- [5] scikit-learn, RandomForestClassifier documentation. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>
- [6] scikit-learn, StratifiedGroupKFold documentation. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html